Lesson-I
TELECOMMUNICATION SWITCHING

Introduction

The world has undergone many changes since the evolution of man. For instance, the exchange of information was initially in the form of signs and sounds. This transitioned to the language and script form with advanced inventions. The communication from one place to another which called for distance between individuals was carried through letters; sent by pigeons and between two groups through drum beats or semaphores. Men used to travel long distances to pass on messages.

Today's world is more an age of communication. The advancement of communication techniques has increased the speed with which the transfer of information takes place. This development has not been an easy process.

At the onset of the invention of communication systems, the invention and usage of telephony was the most important one. The way the telephone systems evolved from a basic system into an essential multi- purpose friendly gadget today, leaves one and all astonished knowing the innovations made out of the meagre resources available in those days.

Telecommunications

The exchange of information between two or many individuals is called **Communication**. The word **tele** is a Greek word which means distance. Hence, **Telecommunication** means the exchange of information between two distant places.

Telecommunications represent the transfer of information, from an entity at one place to an entity at another place, whereas the information can be in the form of data, voice or symbol. The entities can be human beings, computers, facsimile machines, telegraphy machines, phones or so on.

In telephone conversation, the one who initiates the call is referred to as the **Calling Subscriber** and the one for whom the call is destined is the **Called Subscriber**. In other cases of information transfer, the communicating entities are known as **Source** and **Destination**, respectively.

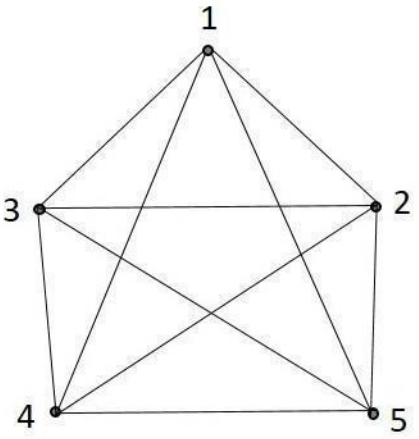
In March 1876, Alexander Graham Bell invented and demonstrated his telephone set and the possibility of long distance voice communication.

He demonstrated the **point-to-point** communication, in which a calling subscriber chooses the appropriate link to establish connection with the called subscriber. This system also requires some mode of **Signalling** to alert the called subscriber about the incoming call and a signal to indicate the calling subscriber, when the called subscriber is busy on another call.

Need for Switching Exchanges

The point-to-point connection for establishing communication requires the telephone sets to be linked using wires. If the number of telephone sets or the subscribers present is low in number, the type of connection will be a little complex. However, if this number is high or moderate, then the connections will lead to a mess. To understand the complication, let us

consider a network of 5 subscribers.The following illustration shows a point-to-point connection for five subscribers (telephone sets):



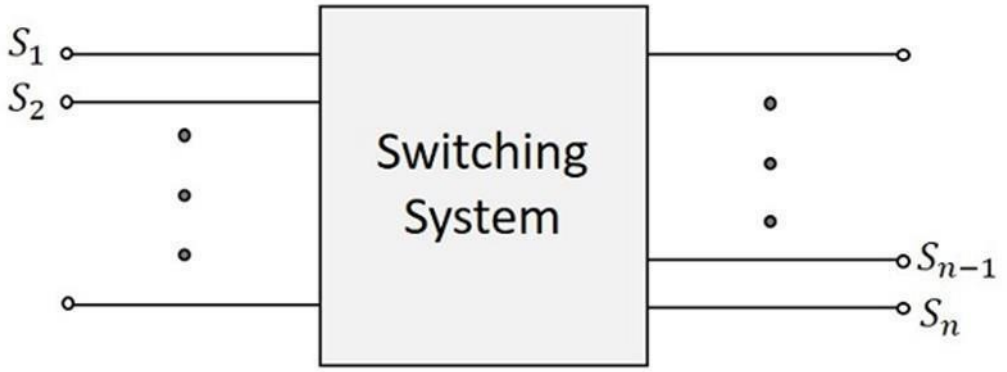
In the point-to-point connection, for **n** entities, we need **n(n-1)/2** links. All these links form a network. Networks with point-to-point links among all the entities are known as **Fully Connected Networks**. The number of links required in a fully connected network becomes very large even with moderate values of **n**.

Hence, a system of switching the networks is needed in-between these subscribers. Alexander Graham Bell recommended the Switching between the subscribers using a switching office that maintains the telephone connections.

Switching Systems

This network connection cannot be simply made with telephone sets and bunch of wires, but a good system is required to make or break a connection. This system is known as the **Switching System** or the **Switching Office** or the **Exchange**. With the introduction of the switching system, the subscribers instead of getting connected directly to one another, are connected to a switching office and then to the required subscriber.

The following figure will help you understand the switching system



With the introduction of switching systems, the need for traditional connections between the subscribers reduced. All the subscribers need to **have a connection with the switching system**, which makes or breaks any connection, requested by the calling subscriber. The switching system, which is also called the **Telephone Exchange**, takes care of establishing the calls. Hence, the total number of such links is equal to the number of subscribers connected to the system.

The early systems required manual operations to establish telephone calls. An operator used to receive a call from the calling subscriber and then connect the call to the called subscriber. Later on, the system was automated.

Telephone Model

The following figure will help you understand the model of telephones in the early stage of its invention.



When you see the telephone in the above figure, the dialer part and the microphone are connected to a stationary wooden plank; and the speaker to listen, was connected by a wire at the side. The top portion of the telephone has two bells connected - these bells ring when there is an incoming call. This is one of the earlier models of the telephone.

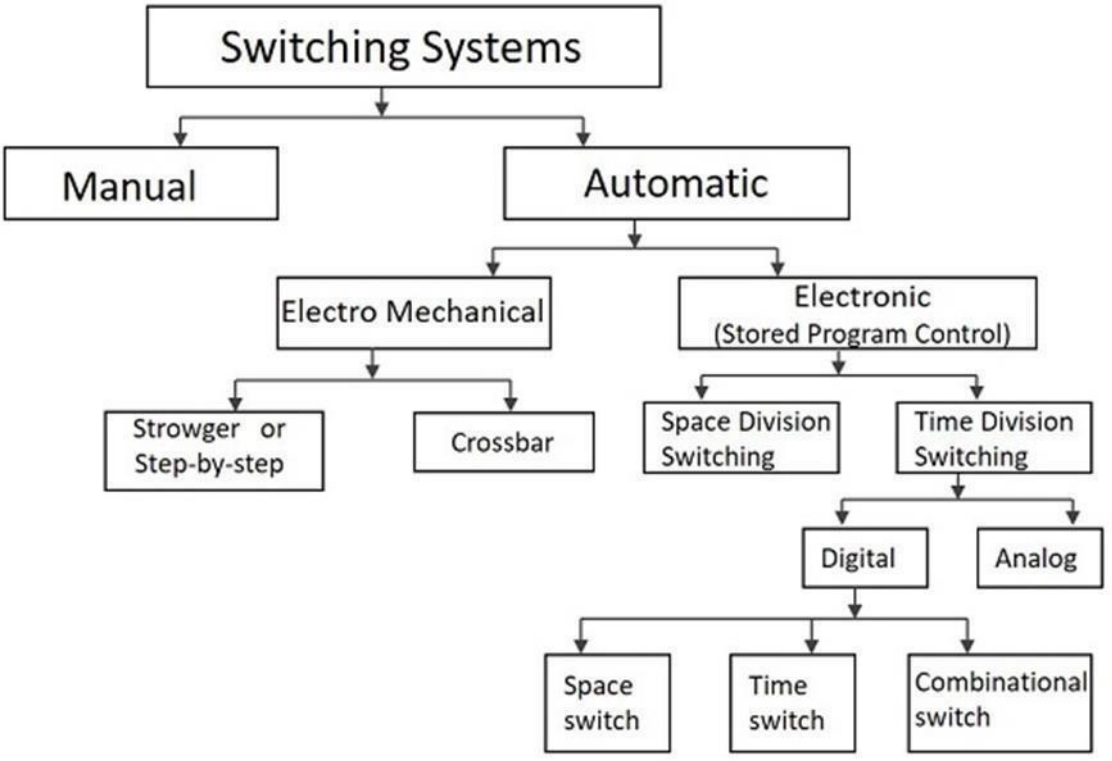
The telephone sets of the calling subscriber and the called subscriber are connected through a switching system or a telephone exchange in order to establish the calls requested.

In the following sections, we will learn about the switching system in detail.

In this chapter, we will understand how the switching systems work. A Switching system can be understood as a collection of switching elements arranged and controlled in such a way as to set up a common path between any two distant points. The introduction of switching systems reduced the complexity of wiring and made the telephony hassle-free.

Classification of Switching Systems

In the early stages of telecommunication systems, the process and stages of switching, played an important to make or break connections. At the initial stages, the switching systems were operated manually. These systems were later automated. The following flowchart shows how the switching systems were classified.



The switching systems in the early stages were operated **manually**. The connections were made by the operators at the telephone exchanges in order to establish a connection. To minimize the disadvantages of manual operation, automatic switching systems were introduced.

The **Automatic** switching systems are classified as the following:

- **Electromechanical Switching Systems** - Here, mechanical switches are electrically operated.
- **Electronic Switching Systems** – Here, the usage of electronic components such as diodes, transistors and ICs are used for the switching purposes.

Electromechanical Switching Systems

The Electromechanical switching systems are a combination of mechanical and electrical

switching types. The electrical circuits and the mechanical relays are deployed in them. The Electromechanical switching systems are further classified into the following.

Step-by-step

The **Step-by-step** switching system is also called the **Strowger** switching system after its inventor A B Strowger. The control functions in a Strowger system are performed by circuits associated with the switching elements in the system.

Crossbar

The **Crossbar** switching systems have hard-wired control subsystems which use relays and latches. These subsystems have limited capability and it is virtually impossible to modify them to provide additional functionalities.

ElectronicSwitching Systems

The Electronic Switching systems are operated with the help of a processor or a computer which control the switching timings. The instructions are programmed and stored on a processor or computer that control the operations. This method of storing the programs on a processor or computer is called the **Stored Program Control** (SPC) technology. New facilities can be added to a **SPC** system by changing the control program.

The switching scheme used by the electronic switching systems may be either **Space Division Switching** or **Time Division Switching**. In space division switching, a dedicated path is established between the calling and the called subscribers for the entire duration of the call. In time division switching, sampled values of speech signals are transferred at fixed intervals.

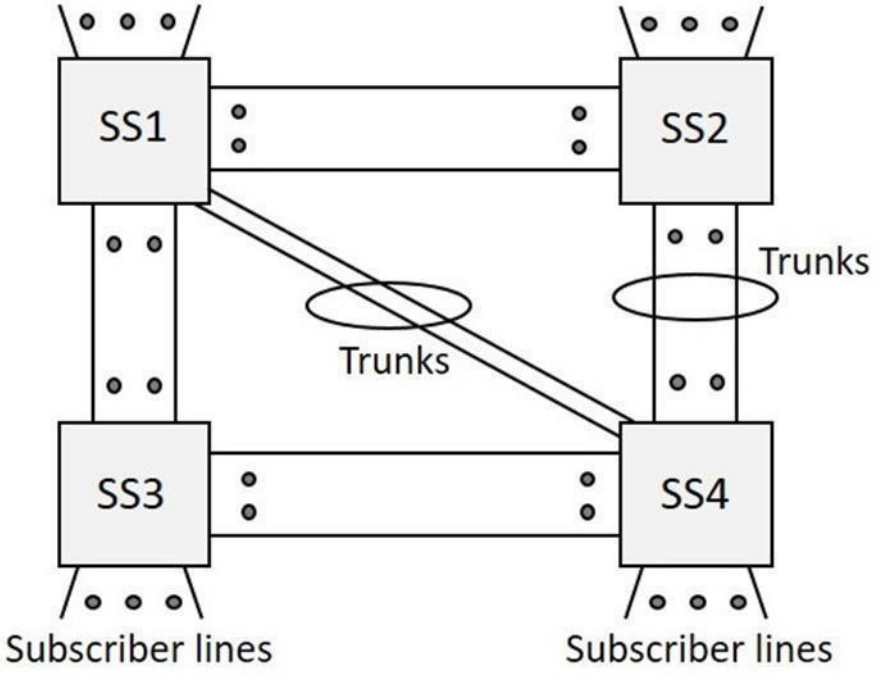
The time division switching may be analog or digital. In analog switching, the sampled voltage levels are transmitted as they are. However, in binary switching, they are binary coded and transmitted. If the coded values are transferred during the same time interval from input to output, the technique is called **Space Switching**. If the values are stored and transferred to the output at a time interval, the technique is called **Time Switching**. A time division digital switch may also be designed by using a combination of space and time switching techniques.

Telecommunication Network

A Telecommunication network is a group of systems that establishes a distant call. The switching systems are part of a telecommunication network.

The switching stations provide connection between different subscribers. Such switching systems can be grouped to form a telecommunication network. The switching systems are connected using lines called the **Trunks**. The lines that run to the Subscriber premises are called the **Subscriber Lines**.

The following figure shows a telecommunication network.



From the early to the later stages of the 20th Century (1900-80), when a person needed to make a distant call, the call was first routed to the operator at the nearest switching center and then the number and location of the called subscriber was noted down. Here, the job of the operator was to establish a call to the remote switching center and then recall the calling subscriber to establish the connection. This system of making calls was called the **Trunk call** system.

For example, a person at Hyderabad can book a trunk call to Mumbai and wait for the operator to call back when the operator establishes connection through the trunk lines and the switching systems.

Communication Links

A telephone switching network is made tip of switching systems trunks. subscriber lines and telephone instruments. Trunks and subscriber lilies are essentially communication links which carry information signals from one point to another. There are basically only two forms of communication links electrical and optical. In the former information is conveyed by means of electrical energy and in the latter. by means of light energy.

The information to be conveyed is not always in the form of electrical or optical signals. For example. human speech signals are essentially sound waves. As a consequence conversion from one form of energy to another may be required before information signals can be carried by communication links. Transducers perform this energy conversion they are available for converting sound. light or heat energy to electrical energy and vice versa.

But at the present state of technological development, there are no known transducers that can directly convert sound energy into light energy. As a result, one is required to go through a two-step process of converting sound into electrical energy first and then into optical energy to be able to use the optical communication links. In other words, today's optical sources accept only electrical signals as input, and the optical detectors produce only electrical signals as output.

Hence, at the transmitting end. The original signals are first converted into electrical signals by using known transducers. Then electrical to optical converters (EOC). i.e. optical sources. are used to obtain optical signals. At the receiving end. optical to electrical converters (OEC). i.e. optical detectors. are used first and then the transducer to reproduce the original signal.

Service Specific Networks

With the concept of switched connections for telephony taking its firm roots. the idea of offering oilier non voice services using switches and switched networks caught the attention of tele communication specialists in the first half of 20th century. Different services nican different types of end equipment's at the customer premises. e.g telex, teleprinter and facsimile machines. The signal characteristics of such end equipment's vary widely.

For example. the electrical characteristics. i.e. voltage. current and power levels and bandwidth of die signal. of a teleprinter are completely different from those of a telephone. In addition. Signalling requirements for different types of end equipments also differ significantly. Such wide variations in electrical characteristics and signalling requirements have led to the development of different service. specific telecommunication networks that operate independently.

Examples are:

1. Telegraph networks

2. Telex networks

3. Telephone networks

4. Data networks

Of the service specific networks mentioned above, telephone networks and data networks form the contents of this text. The most stupendous telecommunications network in existence today is the public switched telephone network (PSTN) or sometimes known as plain old telephone system (POTS). In standards documents.

PSTN is often referred to as general switched telephone network (GSTN). In this text. the popular nomenclature. PSTN is used. PSTN has evolved over a period of 120 years since the beginning of telephony in 1879. There are over a billion telephones in the world connected via land lines (copper cables) to the network. PSTN is highly demanding in its requirements for reliability and availability.

SIMPLE TELEPHONE COMMUNICATION

In the simplest form of a telephone circuit. there is a one-way communication involving two entities. one receiving (listening) and the other transmitting (talking). This form of one-way communication shown in shown as simplex communication. The microphone and the earphone are the transducer

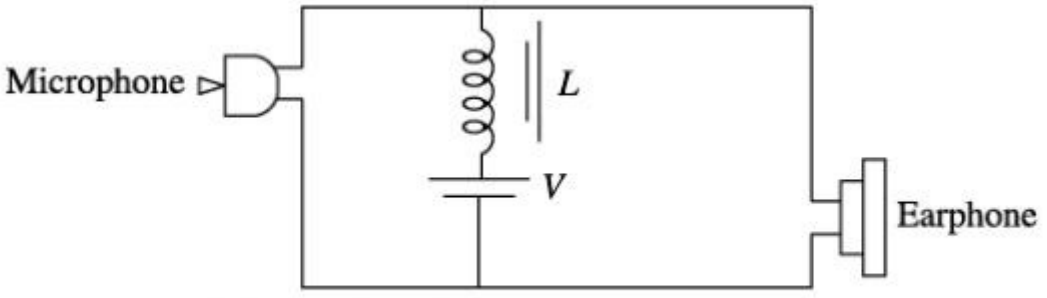


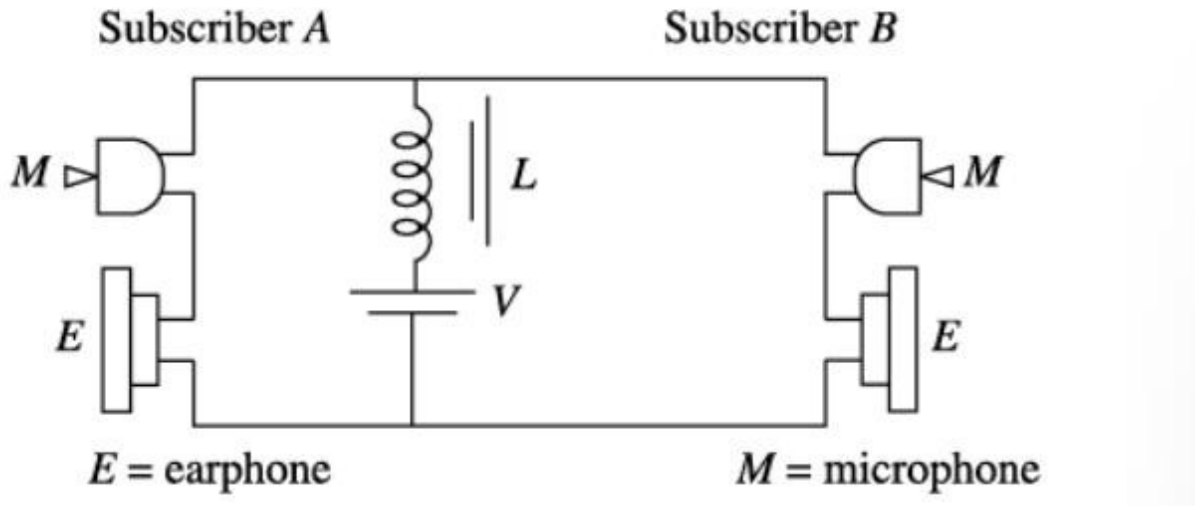
Figure 1.6 A Simplex telephone circuit.

elements of the telephone communication system. Microphone converts speech signals into electrical signals and the earphone converts electrical signals into audio signals. Most commonly used microphone is a carbon microphone. Carbon microphones do not produce high fidelity signals but give out strong electrical signals at acceptable quality levels for telephone conversation.

In a normal telephone communication system. Information is transferred both ways. An entity is capable of both receiving and sending although these do not take place simultaneously. An entity is either receiving or sending at any instant of time. When one entity is transmitting the other is receiving and vice versa. Such a form of communication where the information transfer takes place both ways but not simultaneously is known as half-duplex communication.

If the information transfer takes place in both directions simultaneously. Then it is called full-duplex communication. may be modified to achieve half-duplex communication by the introduction of a transmitter and receiver at both ends of the circuit as shown in Figure. In this circuit. the speech of A is heard by B as well as in A's own earphone.

This audio signal. heard at the generating end. is called side tone. A certain amount of side tone is useful. or even essential. Human speech and hearing system is a feedback system in which the volume of speech is automatically adjusted. based on the side tone heard by the ear. If no side tone is present. a person tends to shout. and if too much of side tone is present. there is a tendency to reduce the speech to a very low level. In the circuit of Figure .The entire speech intensity is heard as side tone. which is not desirable.



BasicsofaSwitching System

In this section, we will learn about the different components and terms used in switching systems.

Inlets and Outlets

The set of input circuits of an exchange are called **Inlets** and the set of output circuits are called the **Outlets**. The primary function of a switching system is to establish an electrical path between a given inlet-outlet pair.

Usually, **N** indicates the inlets and the outlets are indicated by **M**. So, a switching network has **N** inlets and **M** outlets.**Switching Matrix**

The hardware used to establish connection between inlets and outlets is called the **Switching Matrix** or the **Switching Network**. This switching network is the group of connections formed in the process of connecting inlets and outlets. Hence, it is different from the telecommunication network mentioned above.

Types of Connections

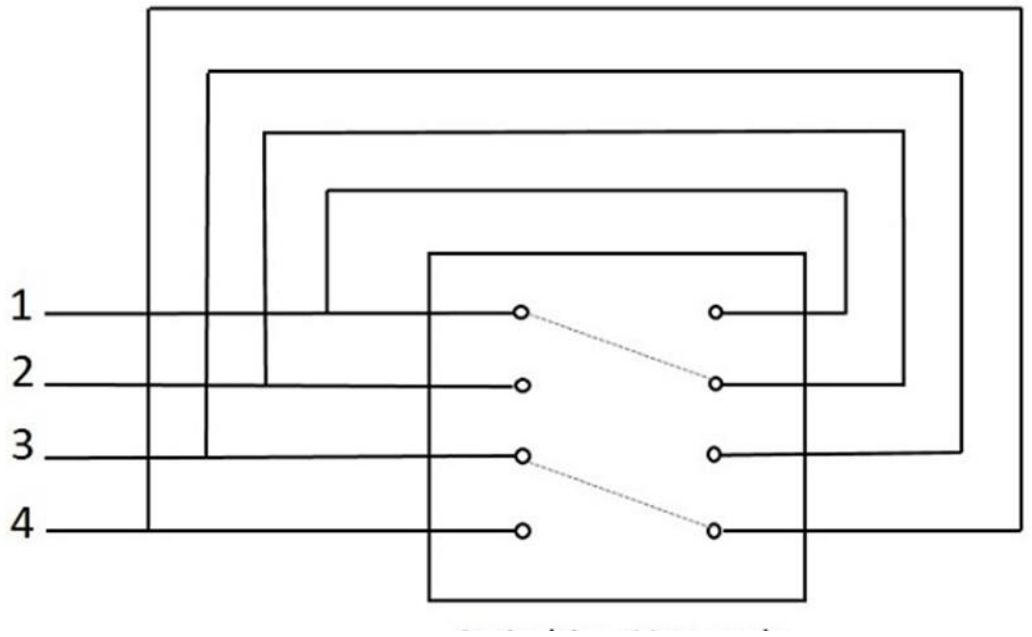
There are four types of connections that can be established in a telecommunication network. The connections are as follows:

- Local call connection between two subscribers in the system.
- Outgoing call connection between a subscriber and an outgoing trunk.
- Incoming call connection between an incoming trunk and a local subscriber.
- Transit call connection between an incoming trunk and an outgoing trunk.

Folded Network

When the number of inlets is equal to the number of outlets for a switching network, such a network is called the **Symmetric Network**, which means N=M. A network where the outlets are connected to the inlets, is called the **Folded Network**.

In a Folded Network, the N number of inlets which come as outlets are again folded back to the inlets. Nevertheless, the switching network provides connections to the inlets and outlets as per the requirement. The following figure will help you understand how the Switching Network works.

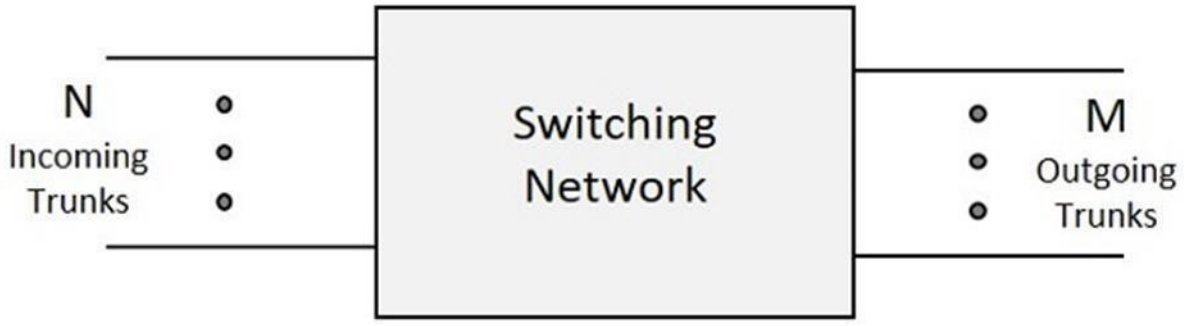


As one connection can be given to one line per time, only N/2 connections are established for N inlets of a folded network. Such a network can be called as **Non-blocking network**. In a non-

blocking network, as long as the called subscriber is free, a calling subscriber will be able to establish a connection to the called subscriber.

In the above figure, only 4 subscribers were considered - where line 1 is busy with line 2 and line 3 is busy with line 4. While the call is in progress, there used to be no chance for making another call and hence, only a single connection was made. Hence for N inlets, only N/2 lines are connected.

At times, it might happen that the inlet and outlet connections are continuously used to make Transit calls through trunk lines only, but not among the local subscribers. The inlet and outlet connections if used in an **Inter-exchange transmission** such that the exchange does not support connection between local subscribers, then it is called the **Transit Exchange**. A switching network of such kind is called the **Non-folded network**. This is shown in the following figure:



BlockingNetwork

If there are no switching paths free in the network, the call requested will be denied, where the subscriber is said to be **blocked** and the network is called the **Blocking Network**. In a blocking network, the number of simultaneous switching paths is less than the maximum number of simultaneous conversations that can take place. The probability that a user may get blocked is called the **Blocking Probability**. A good design should ensure low blocking probability.

Traffc

The product of the calling rate and the average holding time is defined as the Traffic Intensity. The continuous sixty-minute period during which the traffic intensity is high is the Busy Hour. When the traffic exceeds the limit to which the switching system is designed, a subscriber experiences blocking.

Erlag

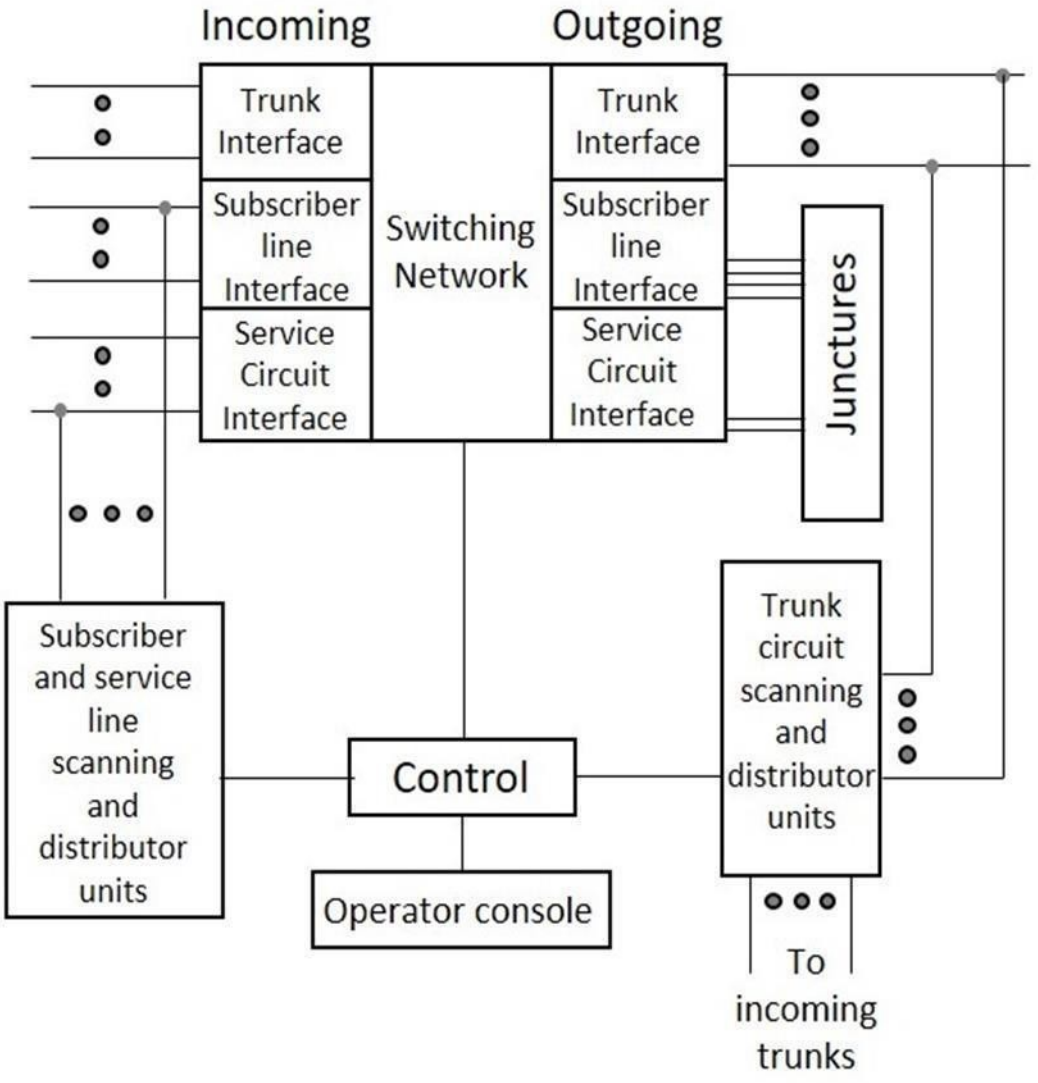
The traffic in a telecommunication network is measured by an internationally accepted unit of traffic intensity known as **Erlang** (E). A switching resource is said to carry one Erlang of traffic if it is continuously occupied through a given period of observation.

Elements of a Switching System

In this chapter, we will discuss the elements of a switching system. Though there are different kinds of switching systems from manual to automatic, a few basic elements play an essential role for the functioning of a switching system. Along with the switching network, there are different sub systems such as control sub system, signaling system, trunk and subscriber line interfaces, distributor units, operator console, juncture circuits, essential for the operation of the whole switching system.

Switching System

In this section, we will understand the structure of the switching system. We will also understand how the different elements work in it. The block diagram of the switching system given below show the essential elements of a switching system.



The diagram shown above contains different blocks of the switching system. The blocks are discussed below.

Switching Network

It provides the switching paths between the called subscribers and the calling subscribers.

Control Subsystem

This is the critical part of the switching system, which actively establishes the switching paths, by identifying the inlet and outlet lines and interpreting the signaling information received on these lines.

This control subsystem, controls the making and breaking of the connection by sensing the signal transfer on the lines. The control sub system sends out signaling information to the subscriber and other exchanges connected to the outgoing trunks.

Signaling

The signaling formats and requirements for the subscriber, the trunks and the sub systems differ significantly. Accordingly, a switching system provides for three different forms of signaling:

- Subscriber loop signaling
- Interexchange signaling
- Intraexchange or register signaling

A switching system is composed of elements that perform switching, control and signaling functions.

Trunk Interface

The Trunk lines used for connections between the switching systems, are terminated at this port. The Trunk interface is the point where the trunk lines are connected to the system.

Subscriber Line Interface

The Subscriber lines used for connections between the subscribers and the switching systems are terminated at this port. The subscriber line interface is the point where the lines from the subscribers are connected to the system.

Line Scanning Unit

The line scanning unit senses and obtains the signaling information from the respective lines. The information obtained from these lines are given to the control sub system to identify the inlets and outlets.

Distributor Units

The distributor units are used for distributing or sending out the signaling information on the respective lines. The distribution of information through the trunk lines, is done through the distribution units

Operator Console

The operator console permits interaction with the switching system for maintenance and administrative purposes.

Service Circuit Interface

The service circuit interface provides interaction between circuits for maintenance and testing purposes.

Junctures

The Junctures is a junction that provides a folded connection for the local subscribers and the service circuits. If the called subscriber and the calling subscriber both are local, then the folded connection helps in making the connection to a local call, whereas the trunk lines will not be in use.

Direct and Indirect

The switching systems are of the following two types:

- the direct control switching system
- the indirect control switching system

Direct Control Switching System

The Switching systems where the control sub systems form an integral part of the network are called the Direct Control Switching systems. For example, the Strowger switching system.

Indirect Control Switching System

The Switching system in which the control sub system is present outside the switching network is called the **Indirect Control** Switching system or the **Common Control** Switching system or the **Register Control** switching system. The examples of this system include Crossbar switching system, Electronic switching system or Stored Program Control method of switching systems.

Strowger Switching System

In this chapter, we will discuss how the Strowger Switching system works. The first ever automatic telephone switching was developed by Almon B Strowger. As the operator at the Manual telephone exchange was the wife of his competitor and was diverting all the business, Strowger thought of developing a switching system, which does not require an operator. This led to the invention of the automatic switching system developed by Strowger.

The **Strowger Switching system** is also called the **step-by-step** switching system as the connections are established in a step-by-step manner.

Automatic Switching System

The Manual Switching system requires an operator who after receiving a request, places a call. Here, the operator is the sole in-charge for establishing or releasing the connections. The privacy of the calls and the details of the called and the calling subscribers are at stake.

Overcoming the disadvantages of Manual Switching systems, the Automatic Switching systems come with the following advantages:

- Language barriers will not affect the request for connection.
- Higher degree of privacy is maintained.
- Faster establishment and release of calls is done.
- Number of calls made in a given period can be increased.
- Calls can be made irrespective of the load on the system or the time of the day.

Let us now throw some light on how a call is made and how dialing is done without the help of an operator.

Dialing

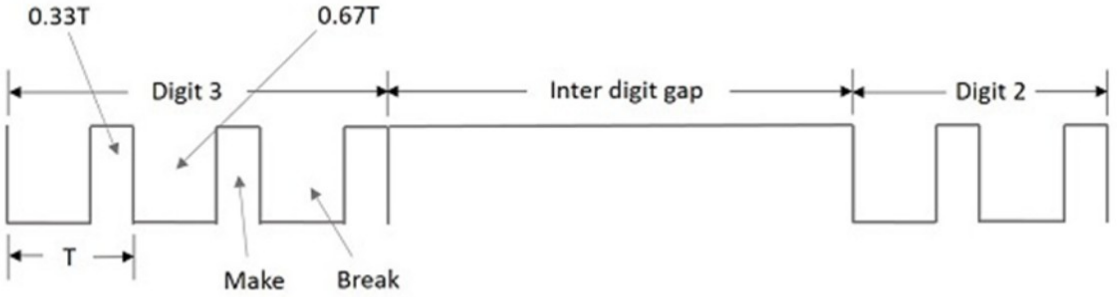
Unlike in Manual Switching system, an automatic switching system requires a formal numbering plan or addressing scheme to identify the subscribers. Numbering plan is where a number identifies a subscriber, is more widely used than the addressing scheme in which a subscriber is identified by the alpha numerical strings. So, there needs to be a mechanism to transmit the identity of the called subscriber to the exchange.

This mechanism should be present in the telephone set, in order to connect the call automatically to the required subscriber. The methods prevalent for this purpose are **Pulse Dialing** and **Multi Frequency Dialing**. Of them, the Pulse dialing is the most commonly used form of dialing till date.

Pulse Dialing

As the name implies, the digits that are used to identify the subscribers are represented by a train of pulses. The number of pulses in a train is equal to the digit value it represents except in the case of zero, which is represented by 10 pulses. Successive digits in a number are represented by a series of pulse trains. These pulses have equal number of time intervals and the number of pulses produced will be according to the number dialed.

Two successive trains are distinguished from one another by a pause in between them, known as the **Inter-digit gap**. The pulses are generated by alternately breaking and making the loop circuit between the subscriber and the exchange. An example pulse train is shown in the following figure.



The above figure shows the pulsating pattern. The pulse rate is usually 10 pulses per second with a 10 percent of tolerance. The gap between the digits, which is called the Inter-digit gap is at least 200ms.

The pulse dialing pattern in recent times employs the duty ratio (ratio between the pulse width and the time period of the waveform) of the pulse as 33 percent nominally and there exists an upper limit for the inter-digit gap.

RotaryDialTelephone

In this section, we will learn about what the Rotary Dial Telephone is and how it works. To start with, we will discuss the drawbacks that were prevalent before the invention of the Rotary Dial Telephone.

The pulse dialing technique is where there is making and breaking of the subscriber loops. This might disturb and affect the performance of speaker, microphone and bell contained in the telephone. In addition, the dialing timings should not affect the timing of the pulse train as this will lead to the dialing of a wrong number.

The Rotary Dial Telephone came into existence to solve the problems prevailing then. The microphone and the loudspeaker are combined and placed in the receiver set. The set has a finger plate the arrangement of which makes the dialing time appropriate.

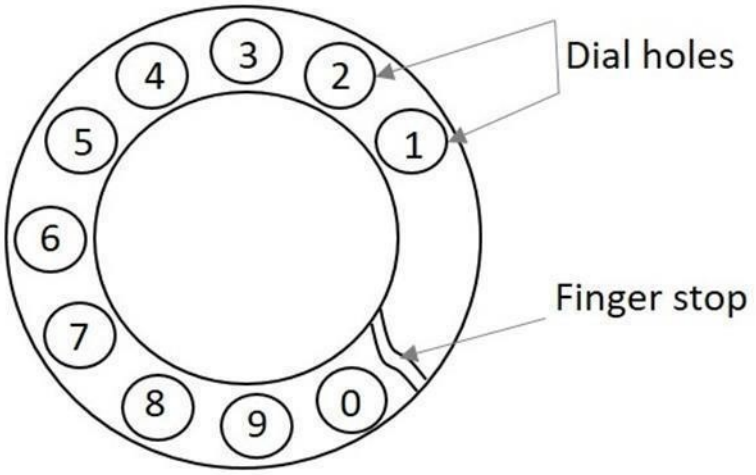
The below figure shows how a rotary dial looks like. The pulse dialing technique is where there is making and breaking of the subscriber loops. This might disturb and affect the performance of speaker, microphone and bell contained in the telephone. In addition, the dialing timings should not affect the timing of the pulse train as this will lead to the dialing of a wrong number.

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The dial is operated by placing the finger in the hole appropriate to the digit to be dialed. Now, drawing the fingerplate round in the clockwise direction to the finger stop position and letting the dial free by withdrawing the finger, makes a number dialed. The fingerplate and the associated mechanism now return to the rest position under the influence of a spring. The dial is ready for the next number.

The dial pulses are produced during the return travel of the fingerplate, thus eliminating the human element in pulse timings. The following figure shows the dial holes and finger stop.



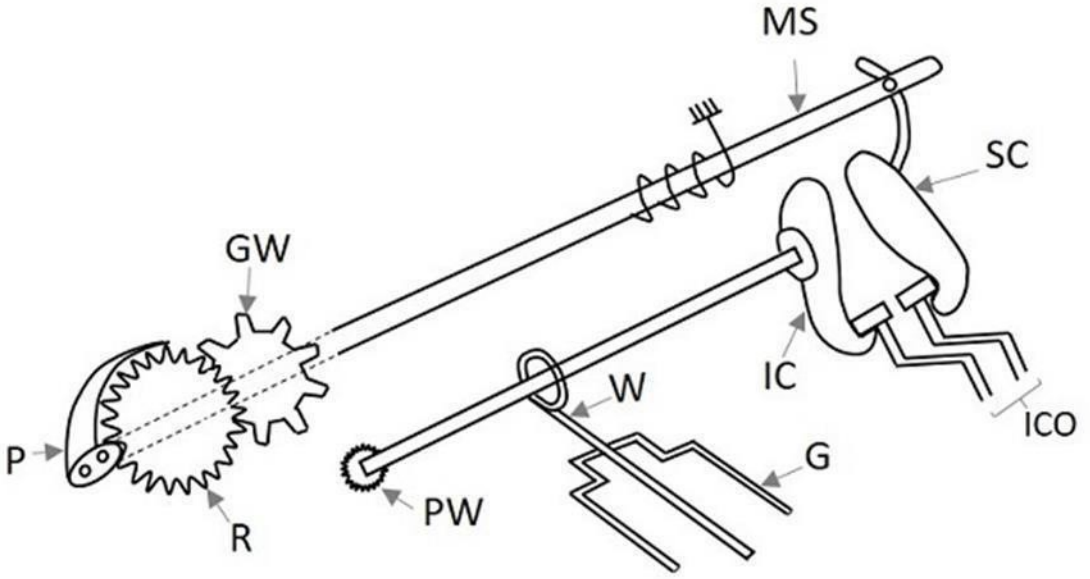
A rotary dial phone uses the following for implementing pulse dialing:

- Finger plate and spring
- Shaft, gear and Pinion wheel

- Pawl and ratchet mechanism
- Impulsing cam and suppressor cam or a trigger mechanism
- Impulsing contact
- Centrifugal governor and worm gear
- Transmitter, Receiver and bell by-pass circuits

Internal Mechanism

The cam mechanism or trigger mechanism helps in dialing. This mechanism is used in operating the Impulsing contact. Let us consider the operation of the rotary dial telephone using the cam mechanism. The following figure will help you understand the internal mechanism.



G = Governo
GW = Gear Wheel
IC = Impulsing Cam
ICO = Impulsing Contacts
MS = Main Shaft

P = Pawl
PW = Pinion Wheel
R = Rachet
SC = Suppressor Cam
W =Worm gear

The suppressor cam helps in keeping the Impulsing cam away from the Impulsing contacts. When the rotary dial is in rest position, then the Impulsing contacts are away from the Impulsing cam. When a number is dialed, by placing the finger in the dial hole, which means the dial is displaced from its position, then the Impulsing contacts come near the Impulsing cam. This rotation of the finger plate, causes the rotation of the Main shaft.

As the dial is rotated in clockwise direction, the pawl slips over the ratchet durin g this clockwise rotation. The ratchet, gear wheel, pinion wheel and the governor are all stationary during the clockwise movement of the dial. When the dial returns, the pawl engages and rotates the ratchet.

All the gear wheel, pinion wheel, the governor rotate, and the uniformity in the speed of the rotation are maintained by the governor. The Impulsing cam, which is attached to a pinion shaft, now breaks and makes the Impulsing contacts that in turn causes the pulses in the circuit.

The shape of the Impulsing cam is such that the break and make periods are in the ratio of 2:1. When the dial is about to reach the rest position, the suppressor cam again, moves the Impulsing contacts away from the Impulsing cam.

This action of getting back to the rest position and waiting for the other number to be dialed creates a gap called the Inter-digit gap, the timing of which is independent of the pause that may occur between two successive digits, due to human dialing habit. This gap is also provided prior to the dialing of the first digit through a small change in the suppressor cam design.

The Pulse generated through this mechanism is then transmitted to the switching systems where the connection to the dialed number is established. The procedure of switching systems is discussed in a subsequent chapter. Meanwhile, let us have an idea on the signaling tones that are used to indicate the condition of the subscribers.

Signaling Tones

In this section, we will understand what are signaling tones and how these work. As the manual exchanges were replaced, the operator who used to communicate the calling subscribers regarding the situation of the called subscribers, needed to be replaced with different tones indicating different situations.

Consider the following five subscriber related signaling functions that are to be performed by the operator:

- Respond to the calling subscriber that system is ready to receive the identification of the called party.
- Inform the calling subscriber that the call is being established.
- Ring the bell of the called party.
- Inform the calling subscriber, if the called party is busy.
- Inform the calling subscriber, if the called party line is unobtainable for some reason.

The function 2 is not signaled in the Strowger switching system. The signaling function 1 is fulfilled by sending a dial tone to the calling subscriber.

Dial Tone

The dial tone is the signaling tone, which indicates that the exchange is ready to accept the

dialed digits from the subscriber. The number should be dialed only when this signal is heard. Otherwise, the digits dialed before this signal will not be considered. This will lead to the dialing of a wrong number.

The dial tone is generally a 33 Hz or 50 Hz or 400 Hz continuous tone as shown below.

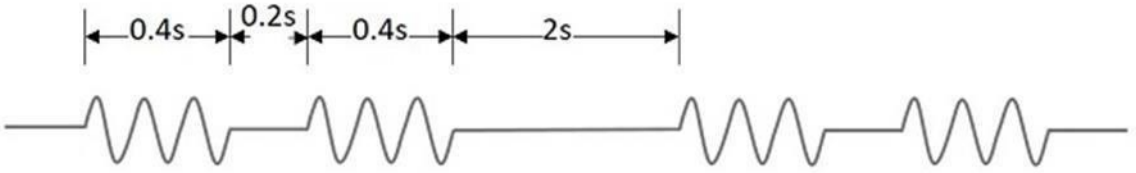


33Hz or 50Hz or 400Hz Continuous signal

Ring Tone

After dialing the number of the called party, when the line of the called party is obtained, the exchange control equipment sends out the ringing current to the telephone set of the called party, which is a familiar double-ring pattern.

Simultaneously, the control equipment sends out a ringing tone to the calling subscriber, which has a pattern similar to that of the ringing current. The two rings double-ring pattern are separated by a time gap of 0.2s and two double-ring patterns by a gap of 2s, as shown in the below figure.



400Hz or 133Hz tone

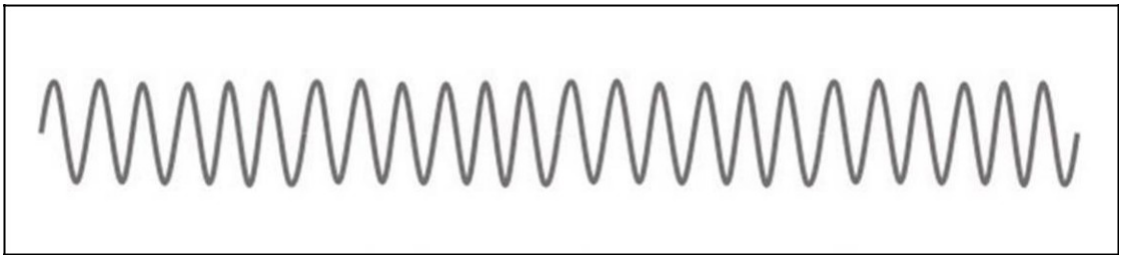
Busy Tone

After dialing the required number, if the called subscriber or the lines at the exchange are not free to place a call, the calling subscriber is sent a busy tone indicating that the lines or the subscriber is busy; this is called a busy tone.

A busty tone of 400Hz signal with silence period in between. The burst and silence durations have the same value of 0.75s or 0.75s.

Number Unobtainable Tone

If the called party is out of order or disconnected or if an error in dialing leads to the selection of a spare line, such a situation is indicated using a continuous 400Hz signal, called as Number Unobtainable tone. The following illustration shows a continuous 400Hz signal.



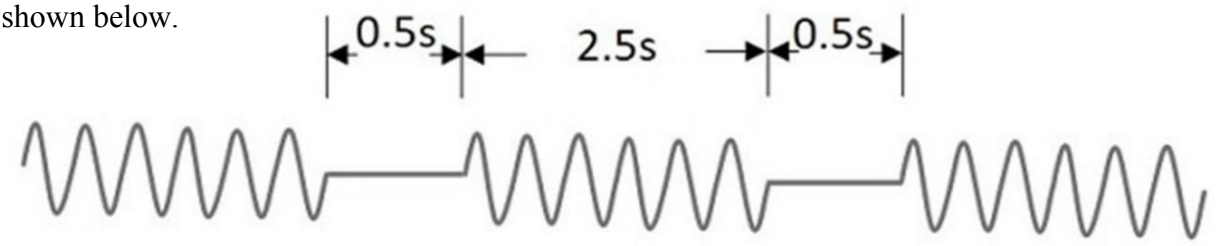
Routing Tone or Call-in-Progress Tone

When a subscriber call is routed through a number of different types of exchanges, one hears different call-in-progress tones as the call progresses through different exchanges. Such a signal is a 400Hz or 800Hz intermittent pattern. This signal has different patterns in different systems.

- In electromechanical systems, it is usually 800Hz with 50 percent duty ratio and 0.5s ON/OFF period.
- In analog electronic exchanges, it is a 400Hz pattern with 0.5s ON period and 2.5s OFF period.
- In digital exchanges, it is 400Hz signal with 0.1s ON/OFF

periods. The signal for routing tone or call-in-progress tone is as

shown below.



In order to overcome the problem of recognizing the difference in these tones for those who are not familiar with telephone signalling and for those who rarely make calls, voice recorded messages were introduced, later on.